

## Learning Unit 1

### 1.1 Types of risks

**Systematic Risk (Market Risk):** Risk that affects the entire market or a specific sector, often beyond an individual's control. Systematic risk is inherent in the market and cannot be eliminated through diversification. It affects all investments to some extent. Examples: Economic downturns, interest rate changes, political instability, and natural disasters.

**Unsystematic Risk (Specific Risk):** Risk specific to a particular company or industry, can often be reduced through diversification. Unsystematic risk is unique to individual investments and can be reduced by holding a diversified portfolio. **Examples:** Company management issues, supply chain disruptions, competition, and regulatory changes.

Key **business risks** include compliance (violations of laws), legal (lawsuits or disputes), strategic (poor decision-making), reputational (damage to brand or image), operational (process failures), human (workforce challenges), security (IT threats), financial (losses from market volatility or credit defaults), competition, and physical risks (natural disasters or theft).

**Managing risk** involves identifying potential risks across all areas, assessing their impact and likelihood, prioritizing critical risks, developing mitigation strategies, and ongoing monitoring to track progress and address new risks. Defining **risk appetite** includes aligning tolerance with objectives, involving stakeholders, fostering a risk-aware culture, and integrating risk tolerance into the organization's strategy.

### 1.2 Financial leverage

Businesses rely on two main sources for financing: debt (liabilities) and equity (owner's capital). Creditors are prioritized over shareholders during bankruptcy, and equity holders receive variable returns (e.g., dividends or share price growth) and non-monetary benefits like voting rights. Debt financing is generally cheaper but riskier for high-debt companies, as banks may start demanding higher interest rates or deny loans.

**Financial leverage** measures the ratio of debt to equity in a company's capital structure. High leverage increases Return on Equity (ROE) but also raises bankruptcy risk and borrowing costs (referred to as Rate of Return Trap). Low leverage offers stability but may lower ROE. There's no universal optimal structure; it varies by company.

- **Low leverage:** financed relatively higher by equity rather than debt > preferred by the bank
- **High leverage:** financed relatively higher by debt rather than equity > preferred by the shareholders

**How debt improves ROE:**

- Cost of debt is normally lower than cost of equity (in high financial leverage can be otherwise because of high risk to creditors because of many loans taken already)
- Interest resulting from debt reduces the total amount of tax paid due to the tax shield
- However, more debt means that more interest has to be paid which affects NI

**To analyse Financial Leverage of the company we can use any of the Solvency ratios**

### 1.3 Dupont Analysis

Used to split ROE into smaller parts and make it more controllable. Usually, Financial leverage has the biggest impact on it. Make sure you check if your Dupont Analysis is correct by comparing to the result you get from the normal ROE formula.

**ROE = ROA \* Financial Leverage ratio**

**ROE = Net Profit Margin \* Asset Turnover \* Financial Leverage ratio**

### 1.4 Asset financing possibilities

According to the fundamental accounting equation, assets can be financed through liabilities or equity, with liabilities typically divided into short-term and long-term. Short-term liabilities are usually the cheapest since they carry less risk for creditors, as they are expected to be repaid within a year. However, they are riskier for the company, as they need to be renewed annually, and interest rates may rise or lenders may refuse further loans. Long-term liabilities, though more expensive due to higher interest rates, are safer for the company as the terms are locked in and repayment is spread over a longer period. Equity is often the most expensive form of financing, as shareholders are last in line to recover their investment if the company faces financial difficulties.

## Learning Unit 2

### 2.1 Financial Statements

A balance sheet, income statement, and cash flow statement are three essential financial statements used by businesses to provide a comprehensive view of their financial performance and position.

#### 1. Balance Sheet:

The balance sheet shows a company's financial position at a specific time, detailing assets (owned), liabilities (owed), and shareholders' equity (residual value). Fundamental Accounting Equation:  $\text{Assets} = \text{Liabilities} + \text{Shareholders' Equity}$

Interconnection: Assets are financed through debt (liabilities) or ownership (equity). Changes in the balance sheet affect other financial statements, e.g., borrowing increases liabilities and appears as cash inflow in the cash flow statement.

## 2. Income Statement (Profit and Loss Statement):

The income statement summarizes revenues, expenses, and net profit or loss over a specific period. Formula:  $\text{Revenue} - \text{Expenses} = \text{Net Income (Profit) or Net Loss}$ . Interconnection: Net income affects shareholders' equity on the balance sheet and cash flow as profit or loss.

## 3. Cash Flow Statement:

The cash flow statement shows cash movement in and out of a company, divided into operating, investing, and financing activities. Formula:  $\text{Operating Cash Flow} + \text{Investing Cash Flow} + \text{Financing Cash Flow} = \text{Change in Cash}$ . Interconnection: It links the balance sheet's cash changes and explains differences between net income and cash flow.

## 2.2 Ratio categories

- **Liquidity:** ability of a hospitality business to meet its short-term obligations
  - **Net Working Capital:**  $\text{Current Assets} - \text{Current liabilities}$
  - **Net Operating Working Capital:**  $\text{NOWC} = \text{AR} + \text{I} + \text{Pre Paid Exp} + \text{any other operating current asset} - \text{AP} - \text{Accrued Expenses} - \text{Advanced Deposits} - \text{any other operating current liability} (*)$

(\*) Does not Include in CA: Cash and Marketable Securities and in the CL: Dividends Payable, Current Part of LTD and Short Term Debt

- **Current ratio:** ratio of total current assets over current liabilities
  - **Current assets/current liabilities**
  - Owners prefer low ratio (less than 1), Creditors prefer high ratio (more than 1)
- **Quick ratio:**  $\text{quick assets (current assets - inventory \& prepaid expenses) / current liabilities}$
- **Working capital turnover:** compares revenue to working capital (CA-CL)
  - **Revenue/ average net working capital** (compared from 2 years)
- **Solvency:** extent to which a company has been financed by debt and if it can meet its long-term obligations
  - **Solvency ratio:**  $\text{total assets / total liabilities}$ 
    - The greater the financial leverage, the lower the ratio
  - **Debt to equity ratio:** compares debt to company's net worth
    - **Total debt/total owners' equity**
  - **Financial Leverage ratio:**  $\text{total assets / total equity}$ 
    - The lower the ratio, the lower the financial leverage
  - **Equity/liabilities:** the higher the ratio, the lower the financial leverage
  - **EBITDA/Interest expenses:** the higher the ratio, the better (shows if we can cover the interest expenses with our EBITDA, more than 1 is good)
  - **Interest coverage ratio (EBIT/Interest expenses):** same as above
- **Activity:** management's effectiveness using its resources
  - **Inventory turnover:** how quickly inventory is being used (how many times a year do we sell all of our inventory)
    - **Cost of sales/average inventory (either take all COS and all inventory or match the specific COS to inventory)**
  - **Paid occupancy %:**  $\text{paid rooms occupied / rooms available}$
  - **Average occupancy per room:**  $\text{\# of guests / rooms occupied}$
  - **Multiple occupancy %:**  $\text{rooms occupied by 2+ people / rooms occupied}$
  - **Seat Turnover:**  $\text{covers served / (number of seats * number of days open)}$
  - **Asset Turnover:**  $\text{total revenue / total assets}$
- **Profitability:** overall effectiveness measured by returns on sales and investments (the rule for all of them -> the higher, the better)
  - **Profit margin:**  $\text{NI / total revenue}$
  - **Operating efficiency:**  $\text{GOP / total revenue}$
  - **EBITDA margin:**  $\text{EBITDA / total revenue}$
  - **Return on assets (ROA):**  $\text{NI / average total assets}$
  - **Return on Equity (ROE):**  $\text{net income / average owner's equity}$
- **Operating:** assesses how well a company performs within its specific industry by evaluating its operational efficiency.
  - **Sales Mix:**  $\text{each department's revenue divided by the total revenue}$
  - **ADR:**  $\text{rooms revenue / number of rooms sold}$
  - **RevPAR:**  $\text{rooms revenue / number of available rooms}$
  - **RevPAC:**  $\text{total revenue / number of guests}$
  - **COS%:**  $\text{COS of the department / department sales}$

## 2.3 Horizontal, vertical and base-year analysis

### 2.3.1 Horizontal analysis

- Comparing results from 2 years with each other
- Relative or absolute numbers  $((\text{new-old})/\text{old})$  and new-old accordingly
- Look at differences in the numbers from the 2 years

### 2.3.2 Vertical analysis (or common-size analysis)

- Compares accounts/revenue from one sheet as a whole
- Compare it to the total (e.g total revenue, Total operating expenses etc.)
- Total always is 100%, rest is parts of the 100%

### 2.3.3 Base-year analysis

- Comparing results of multiple years to one year in order to see if a change in one year was a one-time occurrence or an ongoing trend
- All accounts in the base year stream are considered 100%
- Calculate differences from compared years to base years (New/base year)

## LU3: Working Capital Management

### 3.1 Net (Operating) Working Capital

Working Capital is just your current assets!

Net Working Capital (NWC) and Net Operating Working Capital (NOWC) are financial metrics used to assess a company's short-term financial health, but they focus on different aspects of a business's finances.

NWC indicates how much money remains after paying short-term obligations, with a positive NWC showing that a company has enough to cover its bills, which is favorable for creditors. NOWC, on the other hand, provides a more specific view by focusing on a company's core operations. It considers only the working capital directly tied to day-to-day activities, excluding items that aren't part of day-to-day operations. For example, cash and marketable securities are excluded because they are considered non-operational assets that don't directly contribute to day-to-day business. Additionally, short-term debts unrelated to core business activities are excluded to focus solely on operational working capital. This helps provide a clearer picture of how effectively a company manages its working capital for ongoing business activities.

- Positive N(O)WC -Requires funding to operate: use>source
- Negative N(O)WC -provides funding to operate: use<source
- A **negative net (operating) working capital** is beneficial as it means that suppliers are financing the operations of the company. When managed properly, **negative net (operating) working capital** could be a way to **fund** your growth in sales with other people's **money in a very cheap manner**.
- **Negative net (operating) working capital** often arises when a business generates cash so quickly that it **can** sell its products to the customer before it **has** to pay its bill to the suppliers
- **Opportunity cost** arises when we have too much money invested in current assets such as inventory or accounts receivable when it could be used for other investments like marketable securities that can generate us more money

### 3.2 Cash Conversion Cycle

- CCC is larger part of Operating cycle (ACP + IHP)
- PDP is subtracted from OC to get CCC

Formulas for CCC:

- Inventory Holding Period (IHP)= average inventory/COS\*365 (the goal is to have it as low as possible)
- Accounts Collection (ACP)=average account receivables/credit sales\*365 (the goal is to have it as low as possible) (liquidity ratio)
- Payment Deferral Period (PDP) = average account payables/COS\*365 (the goal is to have it as high as possible)

Turnover is how many times does a certain action happen a year.

- Payable turnover= purchases/average accounts payable (the goal is to have it as low as possible)
- Receivables turnover=revenue/average accounts receivable (the goal is to have it as high as possible) (liquidity ratio)
- Inventory turnover= COS/average inventory (the goal is to have it as high as possible)

Turnover\*period= 365 **ALWAYS**

Period = 365 / turnover

Turnover = 365 / days

**The longer the CCC, the more additional capital is required to finance it**

**The shorter the CCC, the lower the company's reliance on external suppliers of finance**

The shorter the cycle, the better it is for the company as it means inventories are moving through the organisation rapidly, trade receivables are being collected quickly and the organisation is taking the maximum credit possible from suppliers.

To shorten the CCC, reduce IHP, ACP, and increase PDP. ACP can be reduced by accepting only cash or decreasing payment terms, and offering discounts for early payments. Increase PDP by negotiating with suppliers, and decrease IHP by ordering smaller quantities or using JIT inventory management.

Risks of low IHP are that we will not be able to meet the sudden high demands, risks of low ACP are that the customers will be angry and walk away, risks of high PDP are that the suppliers will not work with us anymore.

Learning Unit 4

4.1 Cash Budgeting

How to prepare the cash budget? With credit card fees:

- 1. Look at how much of your sales are paid in cash -> straight revenue (might have to calculate by using a percentage)
- 2. Look at the percentage of credit sales and the delay on the credit sales
- 3. **Watch out for a credit card fee (credit charge) and take it off the credit sales as shown below**
- 4. Outcome of collected accounts:
  - a.  $(\text{sales this month} * \% \text{ of credit sales}) * (1 - \text{credit charge}) * (\text{days in month} - \text{credit card delay}) / \text{days in month}$
  - b.  $(\text{sales last month} * \% \text{ of credit sales}) * (1 - \text{credit charge}) * (\text{delay} / \text{days in the previous month}) = \text{previous month delay sales that we get this month}$
- 5. At the end, the money you receive during the month is: cash sales, sales on account after the credit charge without the delay, sales on account coming from the previous month into the current month
- 6. You may also have some expenses that you have to deduct after all the incoming money or a starting amount of money that you just have to add

| Cash Budget                                 | Sales                     | December           | January      | February     | March        |
|---------------------------------------------|---------------------------|--------------------|--------------|--------------|--------------|
|                                             |                           | 100 000            | 100 000      | 100 000      | 100 000      |
| <b>VARIABLES</b>                            | on account                | 70%                | 70%          | 70%          | 70%          |
| to calculate total cash received from sales | Credit charge             | 4%                 | 4%           | 4%           | 4%           |
|                                             | delay                     | 7                  | 7            | 7            | 7            |
|                                             | days in month             | 31                 | 31           | 29           | 31           |
| <b>OUTCOME</b>                              | <b>Accounts collected</b> |                    |              |              |              |
|                                             | from same month           |                    | 52026        | 50979        | 52026        |
|                                             | from previous month       |                    | 15174        | 15174        | 16221        |
|                                             | <b>Total</b>              | <b>from credit</b> | <b>67200</b> | <b>66154</b> | <b>68246</b> |
|                                             | Cash                      | <b>direct cash</b> | 30000        | 30000        | 30000        |
|                                             | <b>Total</b>              | <b>Total</b>       | <b>97200</b> | <b>96154</b> | <b>98246</b> |

4.2 Cash Flow Statement

4.2.1 Benefits from cash flow statements

A cash flow statement is crucial for management to ensure sufficient working capital for daily operations, support expansion, meet long-term debt obligations, and set dividend policies. For creditors, it helps assess a company’s ability to pay debts and react quickly if cash flow declines. Investors rely on it to determine dividend potential, evaluate expansion opportunities, and assess company valuation based on cash flow.

4.2.2 How to determine if the company is in start up/growth/mature/declining stage:

Every year tells you a new story, therefore you have to look at them as a separate “picture”.

- If operating cash flow is negative = start up phase (we have more expenses than revenue)
- If operating cash flow is a bigger number compared to the first year = growing
- If operating cash flow is a smaller number compared to the first year = declining
- If operating cash flow is stable (both years are quite similar) = mature
- If financing cash flow is positive in both years or even getting more positive = start up/growing (took new loans out, issued more shares)
- If financing cash flow is negative in both years or even getting more negative = declining (paying back our loans, buying back the shares, paying dividends)
- If investing cash flow is positive in both years = declining (selling our non-currents assets, getting rid of them)
- If investing cash flow is negative in both years = start up/growth (buying non-currents assets to use for our business)

## 5.1 Value creation: Road to value

An investment: investment with economic benefits lasting for more than 1 year, for example opening a new factory, entering a new market, acquiring another business or research and development of new products.

### What makes a good investment?:

- Making decisions that maximise the present value or the wealth of a company's owners

### Major factors determining owner's value:

- The **timing** of cash flows: when will they be paid
- **Size** of the cash flows: how much will be paid
- **Riskiness** of the cash flows: how likely is it they will actually be paid

### Profit and value are not the same: value is risk adjusted profit

#### 5.1.2 Impact of financial and operational risk on new investments:

Due to difference in risk; the higher the risk, the higher required rate of return

**Operating risk:** business risk factors that threaten OCF (seasonality, adverse market conditions, competition) related to company's cost structure.

**Financial risk:** risk of not being able to cover liabilities. The higher the risk, the higher required return expected from owners for their investments. A riskier company would receive a higher interest rate on loans.

#### Weighted average cost of capital:

Normally, you need capital for an investment via both debt and equity as one party is not willing to give the full amount.

Both ways come with different costs:

- Cost of debt ( $K_d$ ) is interest charged on a loan
- Cost of Equity ( $R_e$  or  $K_e$ ) is the required return from the shareholders on the invested money
- To satisfy both the bank and shareholders we offer them WACC:

$$WACC = \left( \frac{E}{D + E} \cdot K_e \right) + \left( \frac{D}{D + E} \cdot K_d \right) \cdot (1 - T_c)$$

- WACC will vary depending on the capital structure of a company.
- The higher the debt to equity ratio, the more interest will weigh.
- With low debt to equity ratio, the higher the impact of cost of equity on WACC
- E and D do not necessarily have to be Euro amounts, they can also be found by using ratios such as Debt to Equity or Solvency (remember fundamental accounting equation)
- $K_e$  can be found with ROE formula
- $K_d$  can be found with interest/loan
- Tax rate% can be found with Tax in Euros / EBT
- If we finance investment with only Equity or only Debt, then the other part of the formula disappears as we have 0 there

## 5.2 Time value of Money (understanding compounding & discounting)

### Step B: Discounting or compounding?

- Finding future value(FV) = compounding
- Finding Present value(PV) = Discounting

Formula for FV:  $FV = PV \cdot (1+i)^n$

Formula for PV:  $PV = FV / (1+i)^n$

FV= Future value amount

PV= Present value amount

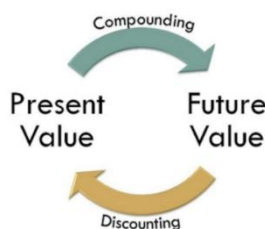
I= interest rate (can be WACC, r, and so on)

N= number of years (interest periods)

Semi-annual compounding:

- Interest rate/2= effective interest rate (no need to adjust the interest rate if it is already semi-annual)
- Period\*2 because of semi annual compounding (think of how many half-a-years will your money grow for)

Same principle applies for quarterly or monthly compounding.



Step C: Determine relevant CF

What to discount or compound?

CF because it is what is actually available

- To the company(to sustain OPS)
- To creditors(interest payments and payback of principal amount)
- To shareholders

Net income has to be adjusted before it can be used to find relevant CF as the **WACC formula includes interest**, the NI has to be recalculated without interest.

The **WACC tells you what return is needed to satisfy the capital providers (shareholders and creditors)**. But Net Income already includes the interest expenses, so using NI would mean that the company already satisfied the creditors!

If you do payback calculation, then interest has to be deducted if you have it (this is done after the EBIT when you do EBIT – interest = EBT and then pay the tax and so on). If you do NPV, PI, IRR or Maximum Price – ignore the interest number even if you are given it!

| Steps         | Years                                        | 1                 | 2    | 3                               |
|---------------|----------------------------------------------|-------------------|------|---------------------------------|
| 1             | Revenues                                     |                   |      |                                 |
| 2             | COS                                          |                   |      |                                 |
| 1-2           | Gross Profit                                 |                   |      |                                 |
| 3             | Fixed costs (ex D, A, and interest expenses) |                   |      |                                 |
| =1-2-3        | EBITDA                                       |                   |      |                                 |
| 4             | Depreciation & Amortisation                  |                   |      |                                 |
| EBITDA - 4    | EBIT                                         |                   |      |                                 |
| !             | Accumulated EBIT                             |                   |      |                                 |
| =EBIT *Tax%   | Tax                                          |                   |      |                                 |
| =EBIT - Taxes | NOPAT                                        |                   |      |                                 |
| 5             | Total CapEx                                  |                   |      |                                 |
| 6             | Change in NOWC                               | Beginning<br>NOWC |      |                                 |
| 4             | Add back depreciation                        |                   |      |                                 |
| =NOPAT-5-6+4  | FCFF                                         |                   |      |                                 |
| 7             | Residual Value                               | -                 | -    |                                 |
|               | Total relevant CF before Terminal Value      | FCFF              | FCFF | FCFF+7                          |
| 8             | Terminal Value                               | -                 | -    | Use the number above to find it |
|               | Yearly Cash Flows Total                      | FCFF              | FCFF | FCFF+7+Terminal value           |
|               | Discounted Total Cash Flows                  |                   |      |                                 |

Terminal value of a project:

- Is the projects value after the end of the projected CF
- Terminal value estimates the value of the company after the period that has been forecasted

It becomes more and more tricky to project cash flows the further in time you go, plus, there is no way to tell how long the company will be in business. To calculate the terminal value of an investment, the **Gordon Growth Model** can be used. This model assumes that the business will continue to generate cash flows at a constant rate for eternity.

Terminal Value= ((CF last year\*(1+g))/(WACC-g)

- CF last year= CF of the last stable forecast year (NOT DISCOUNTED CF) Project CF
- G= terminal growth rate (often long-term inflation rate)
- WACC= weighted average cost of capital

**Terminal Growth Rate** is the constant rate at which the company is forecasted to grow at forever. This growth rate starts at the end of the last forecasted CF. It is usually in line with long-term inflation, yet not higher than the GDP growth rate.



**DON'T forget to discount the terminal value!** You need to discount the terminal value with the last forecasted year; e.g. if you took the FCFF of year 10, you need to discount the terminal value with  $t=10$ .

### 5.3 Capital budgeting models

#### Ranking investments using CBMs:

- In general NPV is preferred as it shows the absolute amount that will be created
- It is always wise to combine NPV with one of the other CBM's to create a clear picture of the situation

|                     |                                                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ARR                 | <ul style="list-style-type: none"> <li>• <b>Accept:</b> if calculated ARR <u>exceeds</u> the required minimum</li> <li>• <b>Reject:</b> if ARR is lower than required minimum</li> </ul>                           |
| Payback             | <ul style="list-style-type: none"> <li>• <b>Accept:</b> if calculated payback period is <u>less</u> than established criterion</li> <li>• <b>Reject:</b> if calculated payback is more than established</li> </ul> |
| NPV                 | <ul style="list-style-type: none"> <li>• <b>Accept:</b> if <math>NPV &gt; 0</math></li> <li>• <b>Reject:</b> <math>NPV &lt; 0</math></li> <li>• <b>Indifferent:</b> <math>NPV = 0</math></li> </ul>                |
| Profitability Index | <ul style="list-style-type: none"> <li>• <b>Accept:</b> if <math>PI &gt; 1</math></li> <li>• <b>Reject:</b> if <math>PI &lt; 1</math></li> <li>• <b>Indifferent:</b> if <math>PI = 1</math></li> </ul>             |
| IRR                 | <ul style="list-style-type: none"> <li>• <b>Accept:</b> <math>IRR &gt; WACC^*</math></li> <li>• <b>Reject:</b> <math>IRR &lt; WACC</math></li> <li>• <b>Indifferent:</b> <math>IRR = WACC</math></li> </ul>        |

*ARR= average annual project income/average investment*

- Average annual project income= Total NI from all years/number of years
- Average investment= (initial investment +scrap value)/2

#### Payback

We use the simple cash flows for payback, meaning we include interest if we are given it.

If the CF are equal every year:

- Payback period= investment cost/annual CF

If the CF are different, we go year by year until we fully cover the investment:

Example:

|        | Cash Flows |
|--------|------------|
| Year 0 | (50.000)   |
| Year 1 | 18.000     |
| Year 2 | 15.000     |
| Year 3 | 14.000     |
| Year 4 | 11.000     |
| Year 5 | 10.000     |
| Year 6 | 9.000      |

Left to cover after Year 1:  $-50\,000 + 18\,000 = -32\,000$ . Left to cover after Year 2:  $-32\,000 + 15\,000 = -17\,000$ . Left to cover after Year 3:  $-17\,000 + 14\,000 = -3\,000$ . Left to cover after Year 4:  $-3\,000 + 11\,000 = 8\,000$  (if we use Year 4 fully we end up covering more than just the investment, therefore we need to find what part of year 4 we need to use exactly)  
 If the leftover Payback amount is smaller than the CF from the next year ( $3\,000 < 11\,000$ ), divide this payback (3 000) by CF from next year:  $3\,000 / 11\,000 = 0.27$

Total payback period = full years used + the part of the next year = 3 full years + 0.27 = 3.27 years

#### Net Present Value:

Calculates whether the stream of future cash flows is worth more than the money we need to invest to buy it or build it. As cash is worth a different amount now than over 10 years we want to calculate the present value of the future cash flows. By discounting the future cash flows of a project you turn the stream of future Euros into today's Euros

- 1) Determine the cash flow (WACC).
- 2) Discount the cash flows with the appropriate discount rate.
- 3) Calculate the sum of all discounted cash flows. This is called the present value.
- 4) Deduct the initial investment from the present value. This answer is your NPV.
- 5) Interpret the NPV. If higher than 0: the investment is worthwhile. If NPV is 0 or lower: the investment does not add value so you do not pursue the project.

*Profitability index(PI) >1 is a go for investment*

$$\text{Profitability Index (PI)} = \frac{\text{Present Value of future cash flows (benefits)}}{\text{Present Value of initial investment (cost)}}$$

The **present value of all future cash flows** represent the benefits of the project. You calculate it the same way as you did for the NPV, namely the sum of all discounted cash flows. If you only have the NPV at hand and not all discounted cashflows, you can simply add

the **initial investment** to the NPV to find the present value of future cash flows. You divide these future benefits by the **present value of the initial investment** (the investment amount).

$$PI \cdot investment = PV \text{ of future cash flows}$$
$$PV \text{ of future cash flows} / PI = investment$$

*Maximum investment:*

We calculate what will the present value of our future cash flows be in order to figure out what is the maximum amount of cash that we can spend on our future in today's terms.

- 1) Determine the cash flow (WACC).
- 2) Discount the cash flows with the appropriate discount rate.
- 3) Calculate the sum of all discounted cash flows. This is called the present value and this is the final answer.

*Internal Rate of Return:*

IRR is the discount percentage (WACC) where  $NPV=0$

IRR= the maximum risk which you can take

IRR depends a lot on the timing of the cash flows. The later we get our bigger cash flows the less risk we can take as they are "punished" more when discounting and hence cannot take a high discount rate. Therefore, projects with bigger cash flows coming early in the timeline usually have higher IRR.

IRR is compared to WACC:

- $WACC > IRR$ , reject the project
- If  $IRR > WACC$ , the project is profitable and should be accepted
- The higher the IRR, the better

To calculate IRR on Excel:

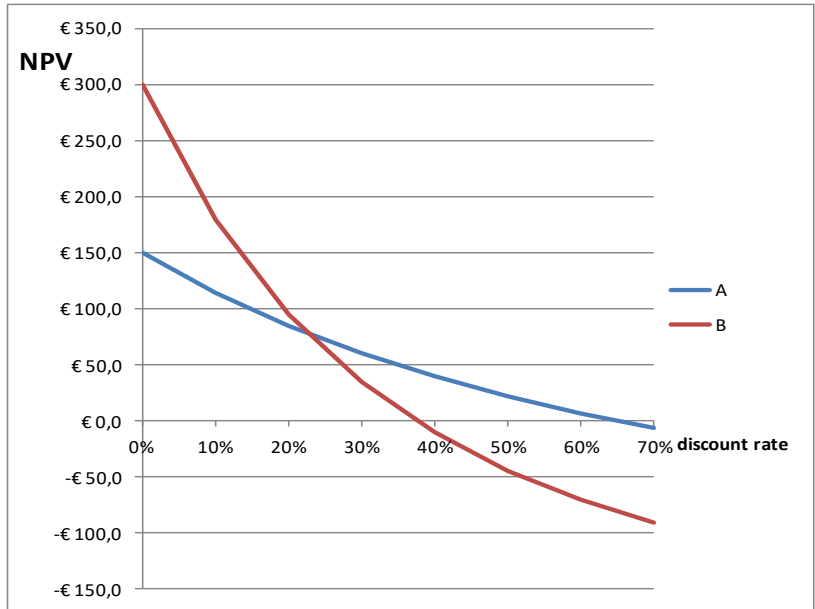
1. Make sure your investment has a negative sign next to it and it is first in the row with the non-discounted cash flows being next to it
2. Go into another cell and start writing  $=IRR$ (select the cells in which your investment and cash flows are)
3. This should give you the most accurate IRR result

To calculate IRR using trial and error method (when given many possible discount rate):

1. With each discount rate that is given find the NPV of the project
2. Whichever discount rate leads to  $NPV=0$  will be the IRR

To find IRR using the graph:

IRR is where the line hits the  $NPV = 0$  line. Project B is approx. 40% and Project A is approx. 65%



**Inflection/indifference point** is the discount rate at which both projects meet. In this case it is approx. 22%.

5.4 Capital Budgeting Methods: advantages and disadvantages

IRR:

- + Considers Time Value of Money
- + Uses cash flows
- + Intuitively easier to understand than NPV
- + Indicates the relative value created (compares it to Investment)
- Lack of scale consideration, ignores size (NPV does the reverse)



- It ignores risk (therefore compare to actual WACC!)
- It is biased in favour of alternatives with quicker payback
- It assumes reinvestment of interim cash flows in projects with equal rates of return

NPV:

- + Incorporates Time value of money by discounting with WACC
- + Uses Cash flows
- + Shows absolute amount of value created
- It ignores the absolute size of the investment needed in its judgement; it shows the absolute value created and not the relative value created (IRR does the reverse)
- Assumes interim cash flows are reinvested at the WACC rate, which might not always be possible

PI:

- + Incorporates Time value of money by discounting with WACC
- + Uses Cash flows
- + Similar to NPV but useful in that it makes it easier to compare two investments (handy when ranking investments)
- Shows only value relative to investment

Payback and ARR:

- + Easy to calculate
- Fails to consider time value of money
- Fails to consider cash flows (only ARR)
- Fails to consider cash flow after payback (only payback)

## Learning Unit 7

In hotel real estate, the business value depends on how well **the hotel operates and the value of the property**. The success of the operations relies on things like the hotel's concept, location, and how good the management is. The value of the property is based on factors such as where it's located, the condition of the building, its size, and the income it makes from renting out rooms or spaces.

Gross Initial Yield (GIY) reflects annual returns before taxes and expenses, whereas Net Initial Yield (NIY) accounts for these deductions.

**GIY = annual return of the investment (before tax\* and expenses) / current price**

**NIY = annual return of the investment (after tax\* and expenses) / current price**

**-Where the current price is the indicative real estate value calculated (formulas can also be moved around to find income or real estate value)**

Real estate valuation normally encompasses room rent, fixed rent, land lease, net owner expenses, and non-recurring capital expenditures (CAPEX). The Loan-to-Value (LTV) ratio compares the amount of the loan to the property's value. Terminal value calculations differ, with "Colliers" using the free cash flow (FCF) of the final forecasted year and an exit multiple based on company value divided by EBITDA, applied to FCF. "Colliers" employs an indexation percentage after a stabilization period, similar to a growth rate (g), and does not include pre-tax expenses in its assessments.

The hotel real estate sector has evolved from owner-operated models to separate entities like "AccorInvest," which owns properties, and "Accor," which manages them. This shift to **"asset-light"** strategies involves "sale and leaseback" deals, enabling capital release. A sale and leaseback is a financial transaction where an owner sells an asset, such as hotel, and then leases it back from the buyer, allowing the original owner to continue using the asset while freeing up capital.

Why not own the hotel itself: owning a hotel isn't always the best option for operators because it allows them to concentrate on their core expertise—delivering top-notch services to guests—without the added burden of property management. By not tying up capital in real estate, they can invest more in expanding their business, enhancing services, or boosting marketing efforts. Additionally, avoiding property ownership reduces financial risks related to market fluctuations and property maintenance. Operators can also benefit from transferring the responsibility of real estate management to professionals, ensuring that each aspect of the business is handled by experts in their respective fields.

Risks of not owning: not owning a hotel comes with risks such as reduced control over property decisions, which can affect the guest experience and adaptability. Additionally, operators face ongoing lease costs and miss the opportunity to build equity and benefit from potential property value appreciation over time.

Hotel real estate is recognized for its specific challenges, including complex structures, varied purposes, diverse risk-return profiles, and market segmentation. Agreements among owners, tenants, management companies, and franchisors differ, affecting investment goals.

Hotel investments are often seen as **riskier** due to cash flow variability, the need for specialized management expertise, and seasonal fluctuations. Market segmentation further differentiates hotel types, with business hotels experiencing different levels of volatility compared to leisure hotels.